

Reactor Design Project

Maleic anhydride from Benzene

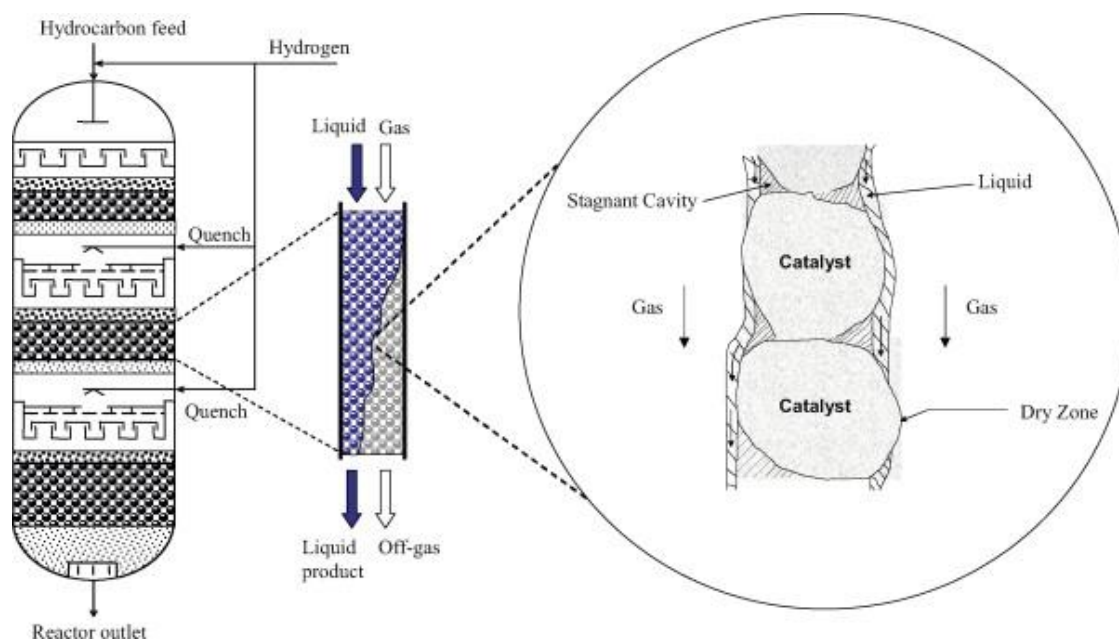
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May 2, 2020

By:

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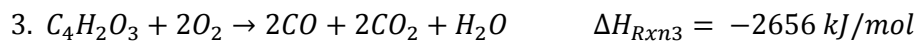
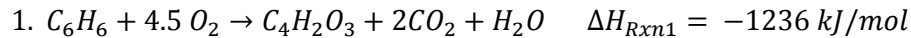


Design Basis:

The reactor to be designed is a packed bed reactor with multiple reaction tubes with external cooling. External co-current cooling is to be provided by a mixed salt flowing at 2000 Kg/h. The tube diameter is to be 25 mm OD to maximize heat transfer (Length = 12 ft and thickness = 1/8"). Heat transfer is a significant issue in this reactor due to the large exothermicity of the oxidation reactions.

The feed (assume physical properties of air) will consist of 15 % benzene in a mixture of oxygen and nitrogen (1:1). Total inlet pressure is 2 atm, and the total molar flow rate is 0.1 kmol/hr (per tube). Pressure drop should be considered using the Ergun equation. The following information may be helpful for the design:

Main reactions:



$$\begin{aligned} r'_1 &= \frac{k_1 P_1^{\alpha_1}}{(1+K_2 P_2^2)} & r'_2 &= k_2 P_1^{\alpha_2} & r'_3 &= \frac{k_3 P_2}{(1+K_2 P_2)^2} \\ k'_1 &= 1E - 6 \frac{\text{kmol}}{\text{kg} \cdot \text{s} \cdot \text{atm}^{0.35}} & k'_2 &= 1.5E - 7 \frac{\text{kmol}}{\text{kg} \cdot \text{s} \cdot \text{atm}^{0.35}} & k'_3 &= 2.9E - 6 \frac{\text{kmol}}{\text{kg} \cdot \text{s} \cdot \text{atm}} \\ K'_2 &= 310 \frac{1}{\text{atm}} & \alpha_1 &= \alpha_2 = 0.35 \end{aligned}$$

From the given reactions it is assumed that the process occurs at steady state. In which, relative rates of reactions are seen equal to each other at equilibrium.

Equation 1

$$\frac{r'_{1A_{Benzene}}}{-1} = \frac{r'_{1A_{Oxygen}}}{-4.5} = \frac{r'_{1A_{MaleicAcid}}}{1} = \frac{r'_{1A_{Carbon Dioxide}}}{2} = \frac{r'_{1A_{Water}}}{1}$$

Equation 2

$$\frac{r'_{2A_{Benzene}}}{-1} = \frac{r'_{2A_{Oxygen}}}{-7.5} = \frac{r'_{2A_{Carbond Dioxide}}}{6} = \frac{r'_{2A_{Water}}}{3}$$

Equation 3

$$\frac{r'_{3A_Maleic\ Acid}}{-1} = \frac{r'_{3A_Oxygen}}{-2} = \frac{r'_{3A_Carbon\ Monoxide}}{2} = \frac{r'_{3A_Carbon\ Dioxide}}{2} = \frac{r'_{3A_Water}}{1}$$

Use of relative rates is useful to determine mole balances for the multiple reactions of flow rates, pressure drops, coolant and reactor temperature as differential equations.

Differential Flow Rates with respect to catalyst weight

$$\frac{dF_{A_Benzene}}{dW} = -r'_{1A} - r'_{2A}$$

$$\frac{dF_{A_Oxygen}}{dW} = -4.5r'_{1A} - 7.5r'_{2A} - 2r'_{3A}$$

$$\frac{dF_{A_Maleic\ Acid}}{dW} = r'_{1A} - r'_{3A}$$

$$\frac{dF_{A_Water}}{dW} = 2r'_{1A} + 3r'_{2A} + 3r'_{3A}$$

$$\frac{dF_{A_Nitrogen\ (Inert)}}{dW} = 0$$

$$\frac{dF_{A_Carbon\ Monoxide}}{dW} = 2r'_{3A}$$

$$\frac{dF_{A_Carbon\ Dioxide}}{dW} = 2r'_{1A} + 6r'_{2A} + 2r'_{3A}$$

Differential Temperatures

Temperature change is a function of varying volume, catalyst weight, and length of the reactor.

The initial temperature differential is modeled on a per component basis of the above reactions

$$\frac{dT}{dV} = \frac{U_{outter}(T_a - T) + \sum[(-r_{i,j})(-\Delta H_{Rxn i,j}(T))]}{\sum[F_{i,j}(C_{pi,j})]}$$

Let catalyst weight be a function of volume

$$dW = \rho_b(dV)$$

And volume a function of length with area as

$$dV = dZ * A_c$$

These relations allow

$$\frac{dT}{dW} = \frac{\frac{U_{outter}(T_a - T)}{\rho_b} + \sum[(-r'_{i,j})(-\Delta H_{Rxn i,j}(T))]}{\sum[F_{i,j}(C_{pi,j})]}$$

$$\frac{dT}{dZ} = A_c * \frac{U_{outter}(T_a - T) + \sum[(-r'_{i,j})(-\Delta H_{Rxn i,j}(T))]}{\sum[F_{i,j}(C_{pi,j})]}$$

Relation to co and counter current flows

In a packed bed reactor, the coolant temperature (in this case salt temperature) may be modeled as co-current or counter current. The model may be based on weight or length parameters.

Simplified for co-current flow in terms of length

$$\frac{dT_a}{dZ} = A_c * \frac{U_{outter}(T - T_a)}{C_{Psalt(T_R)}(\dot{m})_{salt}}$$

Simplified for counter-current flow in terms of length

$$\frac{dT_a}{dZ} = A_c * \frac{U_{outter}(T_a - T)}{C_{Psalt(T_R)}(\dot{m})_{salt}}$$

Simplified for co-current flow in terms of weight

$$\frac{dT_a}{dW} = \frac{U_{outter}(T - T_a)}{\rho_b(\dot{m}_{salt})C_{Psalt}}$$

Simplified for counter-current flow in terms of weight

$$\frac{dT_a}{dW} = \frac{U_{outter}(T_a - T)}{\rho_b(\dot{m}_{salt})C_{Psalt}}$$

Differential Pressure Drop

The equation of Ergun pressure drop is used to model the differential pressure drop in relation to volumetric flow rate, initial temperature, reactor temperature and flow rates.

Certain parameters are modeled in terms of Dalton's law of partial pressures for the reactant Benzene to the product Maleic Acid.

$$P_{Benzene} = \frac{F_{A_Benzene}}{F_{To}} * P_{Total}$$

$$P_{Maleic\ Acid} = \frac{F_{A_Maleic\ Acid}}{F_{To}} * P_{Total}$$

Where

F_{To} = Overall inlet flow rate

P_{Total} = Inlet total pressure

$F_{A_Benzene}$ = Flow rate of Benzene

$F_{A_Maleic\ Acid}$ = Flow rate of Maleic Acid

In Which the pressure drop may now be modeled using the Ergun equation

$$\frac{dP}{dW} = - \frac{\beta_o}{A_c(1 - \phi)\rho_c} \frac{P_o}{P} \left(\frac{T}{T_o} \right) \frac{F_T}{F_{To}}$$

Where,

$$\beta_o = \frac{G(1 - \phi)}{\rho_o * g_c * D_p * \phi^3} * \left[\frac{150(1 - \phi)\mu}{D_p} + 1.75G \right]$$

ϕ = void fraction

g_c = 1 (in SI units)

D_p = outer diameter of tube

P_o = initial pressure

P = final pressure

T = temperature reactor exit

A_c = cross sectional area of tube

μ = viscosity as a function of temperature

G = as a function of density and superficial mass velocity

ρ_o = initial density of air as a function of temperature

G as a function of density and superficial mass velocity

$$G = \rho * u$$

$$u = \frac{v_o}{A_c}$$

v_o = Inlet volumetric flow rate

Where the assumption of an ideal allows v_o to be found as

$$P_o(v_o) = (F_{To})(R)(T_o)$$

$$v_o = \frac{F_{To}(R)T_o}{P_o}$$

The initial calculation of the volumetric flow rate is only valid at the inlet and changes as a functions of temperature, flow rates, and pressure.

Relation of volumetric flow rate to variational parameters

$$\frac{v}{v_o} = \frac{F_T}{F_{To}} \left(\frac{T}{T_o} \right) \frac{P_o}{P}$$

In A similar manner the density of the gas also varies and may be accounted for as a closed system mass balance, “in = out”.

$$\dot{m}_o = \dot{m}$$

$$\rho_o(v_o) = \rho(v)$$

$$v_o = \frac{F_{To}(R)T_o}{P_o} = v = \frac{F_T(R)T}{P}$$

$$\frac{v}{v_o} = \frac{F_T}{F_{To}} * \frac{T}{T_o} * \frac{P_o}{P}$$

After substituting the relation of $\frac{v}{v_o}$ as v_o & v back into the mass balance a relation for density is obtained as

$$\rho = \rho_o * \frac{P}{P_o} * \frac{T_o}{T} * \frac{F_{To}}{F_T}$$

Viscosity as a function of temperature

As viscosity changes with temperature the reference temperature is set at 273.15 k with a

$$\mu(T) = (\mu_{T_o}) \left(\frac{T}{273.15} \right)^{1.5} \left(\frac{273.15 + 110}{T + 110} \right)$$

μ_{T_o} = reference viscosity

Accounting for K at different Temperatures:

Arrhenius Equation

$$k_j(T) = k_j(T_o) * \exp \left[\frac{E_j}{R} * \left(\frac{1}{T_o} - \frac{1}{T} \right) \right]$$

$$k_1(T) = 1.77E - 5 * \exp \left[\frac{83100 \frac{J}{mol}}{8.314 \frac{J}{mol \cdot K}} * \left(\frac{1}{673.15 K} - \frac{1}{T} \right) \right]$$

$$k_2(T) = 2.66E - 6 * \exp \left[\frac{83100 \frac{J}{mol}}{8.314 \frac{J}{mol \cdot K}} * \left(\frac{1}{673.15 K} - \frac{1}{T} \right) \right]$$

$$k_2(T) = 2.86E - 8 * \exp \left[\frac{155000 \frac{J}{mol}}{8.314 \frac{J}{mol \cdot K}} * \left(\frac{1}{673.15 K} - \frac{1}{T} \right) \right]$$

* K units $\frac{mol}{kg \cdot s} * \frac{m^2}{N}$

Van Hoff

$$K_i(T) = K_i(T_o) * \exp \left[\frac{\Delta H_{RXi}^o(T_R) - T_R \Delta C_{Pi}}{R} * \left(\frac{1}{T_o} - \frac{1}{T} \right) + \frac{\Delta C_{Pi}}{R} \ln \frac{T}{T_o} \right]$$

$$K_2(T) = 3.05E - 3 * \exp \left[\frac{-2,656,000 \frac{J}{mol} - 573.15K(\Delta C_{P2})}{R} \left(\frac{1}{673.15K} - \frac{1}{T} \right) + \frac{\Delta C_{P2}}{R} \ln \frac{T}{673.15K} \right]$$

*Where ΔC_{P2} is calculated via Shomate equation.

Shomate equation

Shomate equation is followed by the overall change of Cp per molecule followed by integrals to solve for Temperature

$$C_{Pi} \left(\frac{kJ}{kmol.K} \right) = A_i + B_i T + C_i T^2 + D_i T^3$$

Constant values available via Nist/Athena database per molecule.

Example for Inlet Salt

$$C_{Psalt} \left(\frac{kJ}{kmol.K} \right) = 50 + (2E - 3)T + (1E - 5)T^2 + (-4E - 9)T^3$$

It is important to note that a change in Specific heat for reaction 1,2 and 3 are:

$$\Delta C_{p_{Rxn1}} = 2(\Delta C_{p_{Carbon\ Dioxide}}) + \Delta C_{p_{Water}} + \Delta C_{p_{Maleic\ Acid}} - \Delta C_{p_{Benzene}} - 4.5(\Delta C_{p_{Oxygen}})$$

$$\Delta C_{p_{Rxn2}} = 6(\Delta C_{p_{Carbon\ Dioxide}}) + 3(\Delta C_{p_{Water}}) - \Delta C_{p_{Benzene}} - 7.5(\Delta C_{p_{Oxygen}})$$

$$\Delta C_{p_{Rxn3}} = 2(\Delta C_{p_{Carbon\ Monoxide}}) + 2(\Delta C_{p_{Carbon\ Dioxide}}) + \Delta C_{p_{Water}} - \Delta C_{p_{Maleic\ Acid}} - 2(\Delta C_{p_{Oxygen}})$$

In which these overall changes are beneficial for integral modeling for relation to the length parameter “Z”.

Reaction 1:

$$\int \Delta C_{p_{Rxn1}} = 2 \int \Delta C_{p_{water}} + \int \Delta C_{p_{Maleic\ Acid}} + 2 \int \Delta C_{p_{Carbon\ Dioxide}} - \int \Delta C_{p_{Benzene}} - 4.5 \int \Delta C_{p_{Oxygen}}$$

Reaction 2:

$$\int \Delta C p_{Rxn2} = 3 \int \Delta C p_{water} + 6 \int \Delta C p_{Carbon\ Dioxide} - \int \Delta C p_{Benzene} - 7.5 \int \Delta C p_{Oxygen}$$

Reaction 3:

$$\int \Delta C p_{Rxn2} = \int \Delta C p_{water} + 2 \int \Delta C p_{Carbon\ Dioxide} + 2 \int \Delta C p_{Carbon\ Monoxide} - \int \Delta C p_{Maleic\ Acid} - 2 \int \Delta C p_{Oxygen}$$

Length parameter

$$Z = -(r'_1)(\Delta H_{Rxn1}) \left(\int \Delta C p_{Rxn1} \right) - (r'_2)(\Delta H_{Rxn2}) \left(\int \Delta C p_{Rxn2} \right) - (r'_3)(\Delta H_{Rxn3}) \left(\int \Delta C p_{Rxn3} \right)$$

Concluding Step

At this point all values are accounted for in the differential equations of temperatures, pressure drop, and flow rates of each component involved in the production of Benzene to Maleic Acid. The proceeding step to model the PBR reactor is using a software such as Polymath or Matlab. These software's will simultaneously solve the differential equations needed. For this PBR a different equation per seven components for flow rate, one pressure differential and two temperature differentials.

Results

Deliverable 1)

Model the maleic anhydride reactor assuming an inlet reactant temperature of 300C and inlet salt temperature of 150C, calculating the conversion of benzene. Plot the pressure, reactor-temperature, salt-temperature and molar flow rates of all the components as functions of reactor length.

Benzene Conversion:

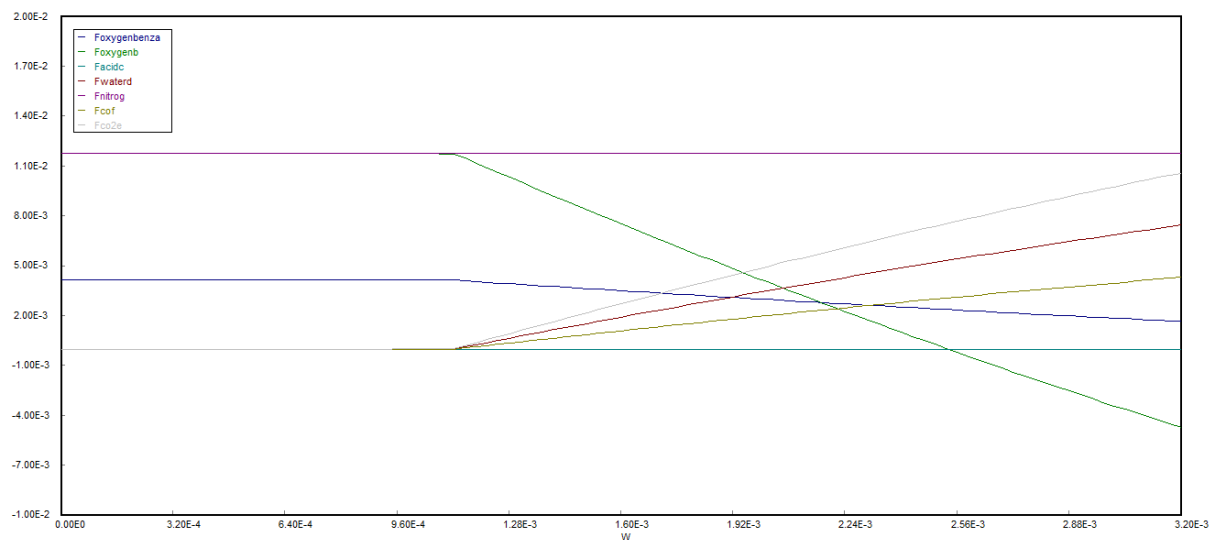
From Appendix ii, the initial flow rate of Benzene entering the PBR tube is 0.004167 mol/s. The outlet flow rate of Benzene at the exit of the PBR tube is 0.0016755 mol/s. Taken to 4 significant figures:

$$X = \frac{F_{Ao} - F_A}{F_{Ao}} * 100 \%$$

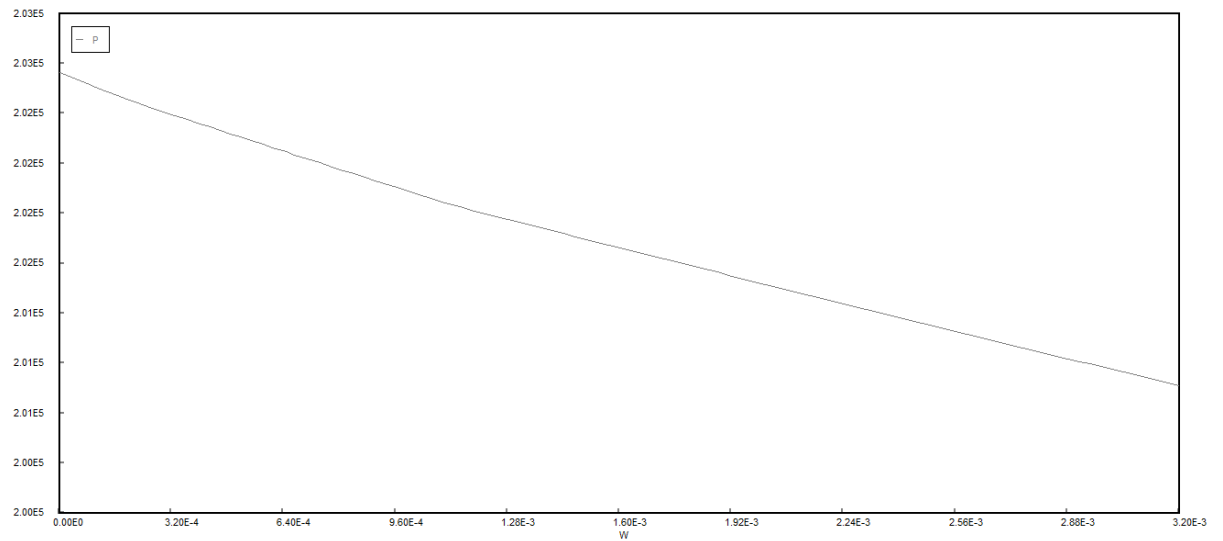
$$X = \frac{0.0042 - 0.0017}{0.0042} * 100 \%$$

$$X = 59.52 \%$$

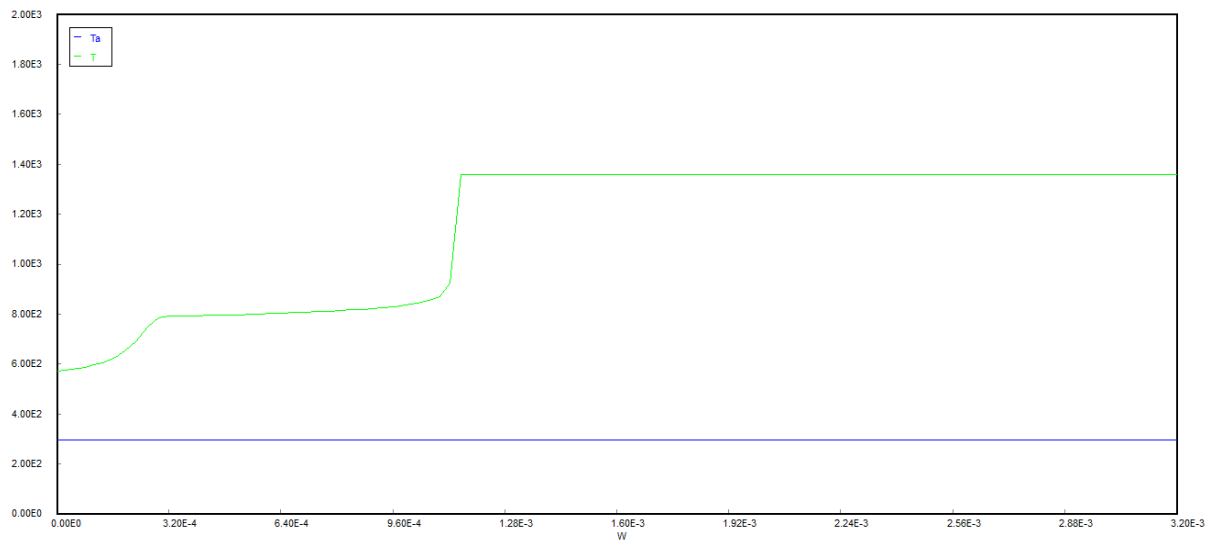
Graph 1: Molar Flow Rates Vs Catalyst Weight (function of Length)



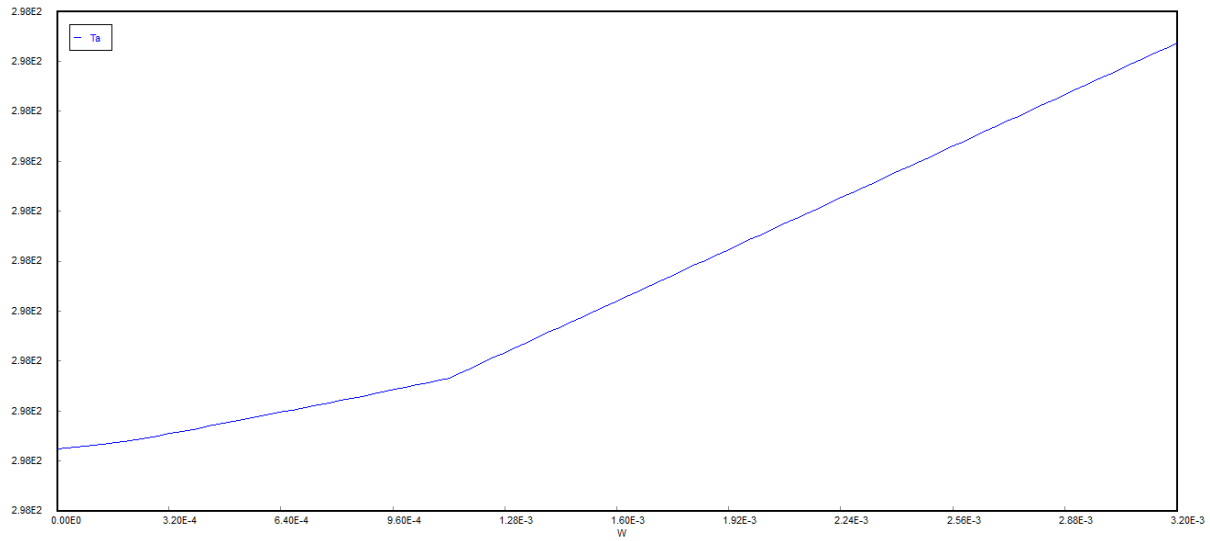
Graph 2: Pressure Vs Catalyst Weight (function of Length)



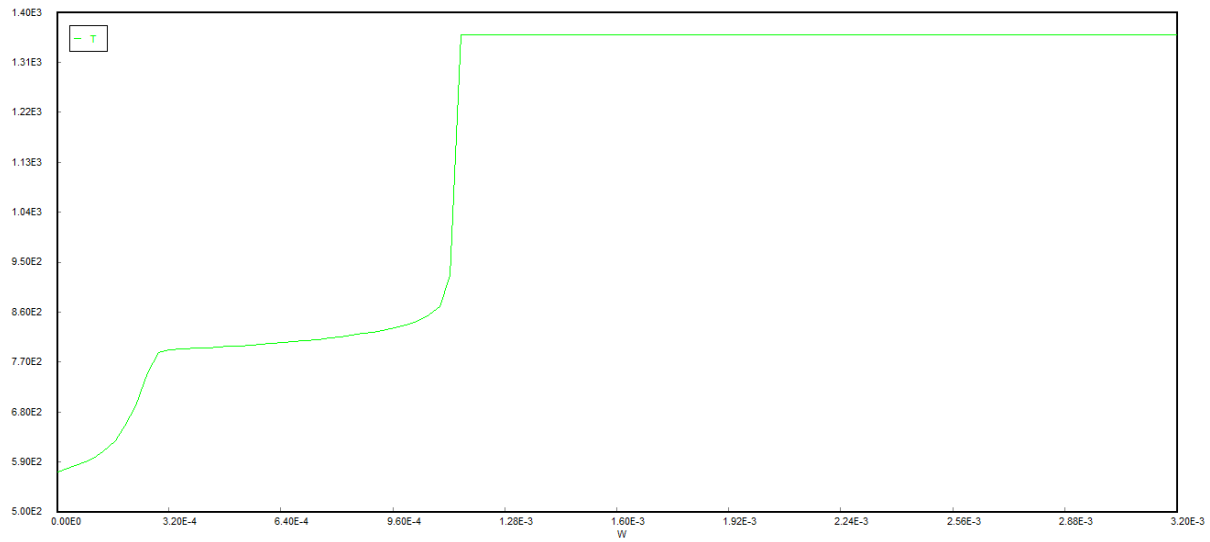
Graph 3: Reactor-temperature & Salt-temperature Vs Catalyst Weight (function of Length)



Graph 3A: Reactor-temperature & Salt-temperature Vs Catalyst Weight (function of Length)



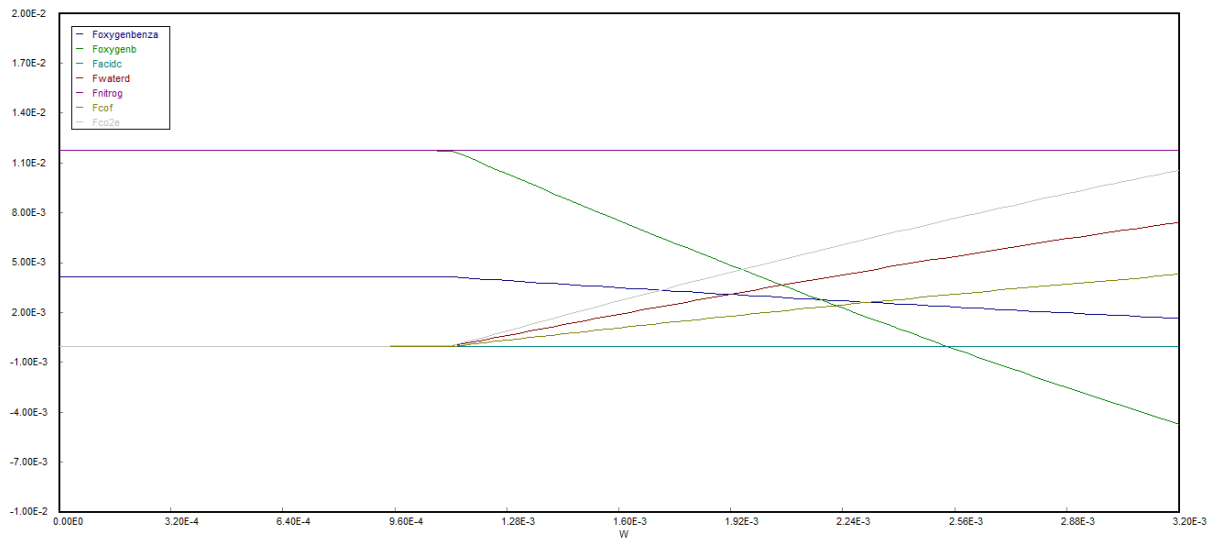
Graph 3B: Reactor-temperature & Salt-temperature Vs Catalyst Weight (function of Length)



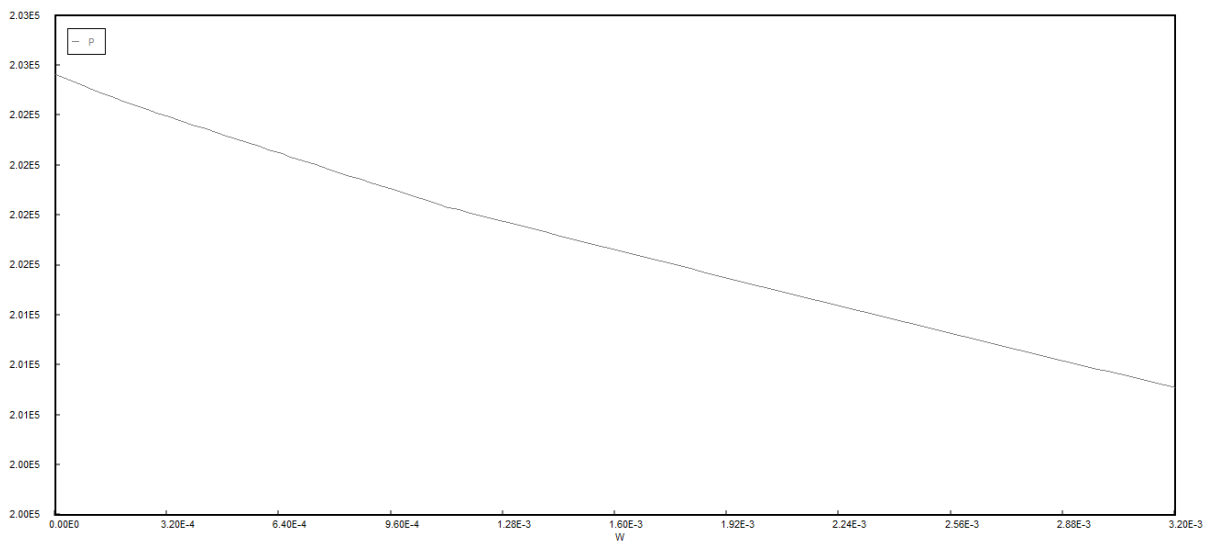
Deliverable 2)

What will be the analysis for part 1 with counter-current heat exchanger?

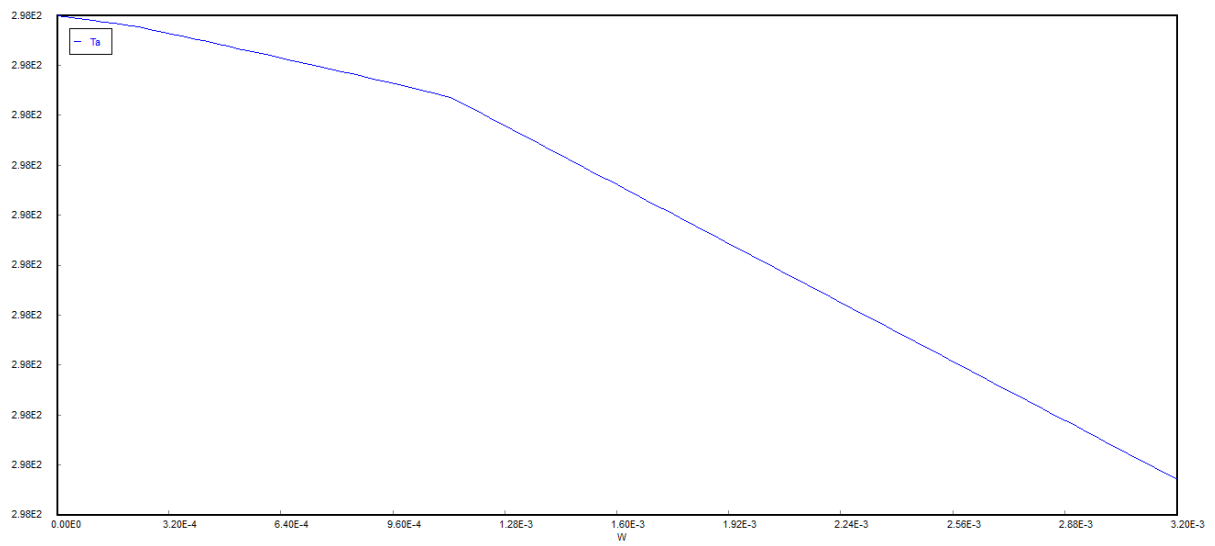
Graph 4: Molar Flow Rates Vs Catalyst Weight (function of Length)



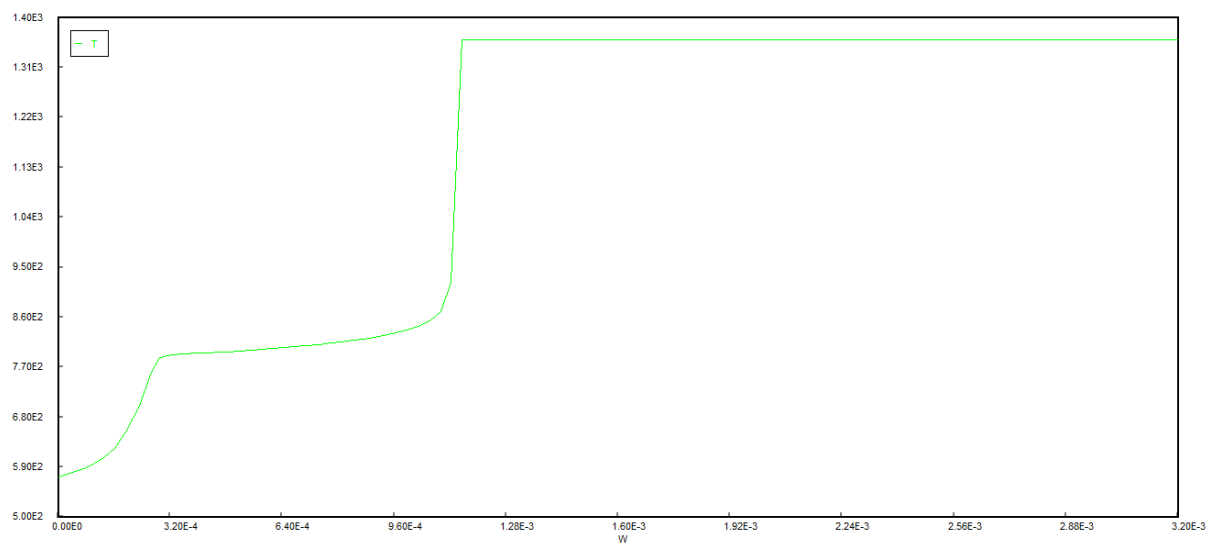
Graph 5: Pressure Vs Catalyst Weight (function of Length)



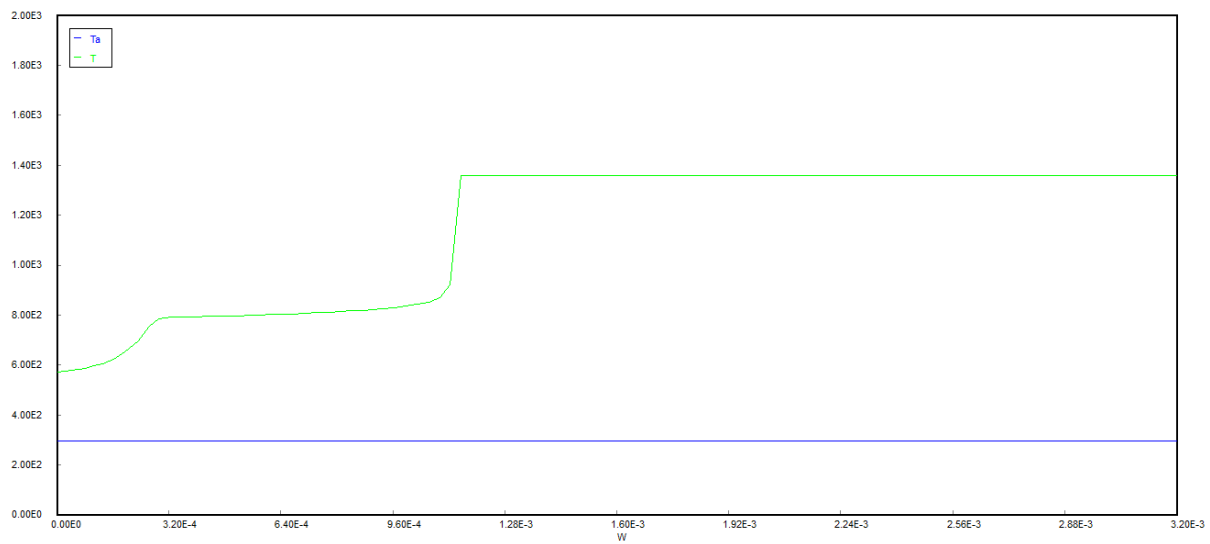
Graph 6: Reactor-temperature & Salt-temperature Vs Catalyst Weight (function of Length)



Graph 6A: Reactor-temperature & Salt-temperature Vs Catalyst Weight (function of Length)



Graph 6B: Reactor-temperature & Salt-temperature Vs Catalyst Weight (function of Length)



Deliverable 1 & 2 Discussion)

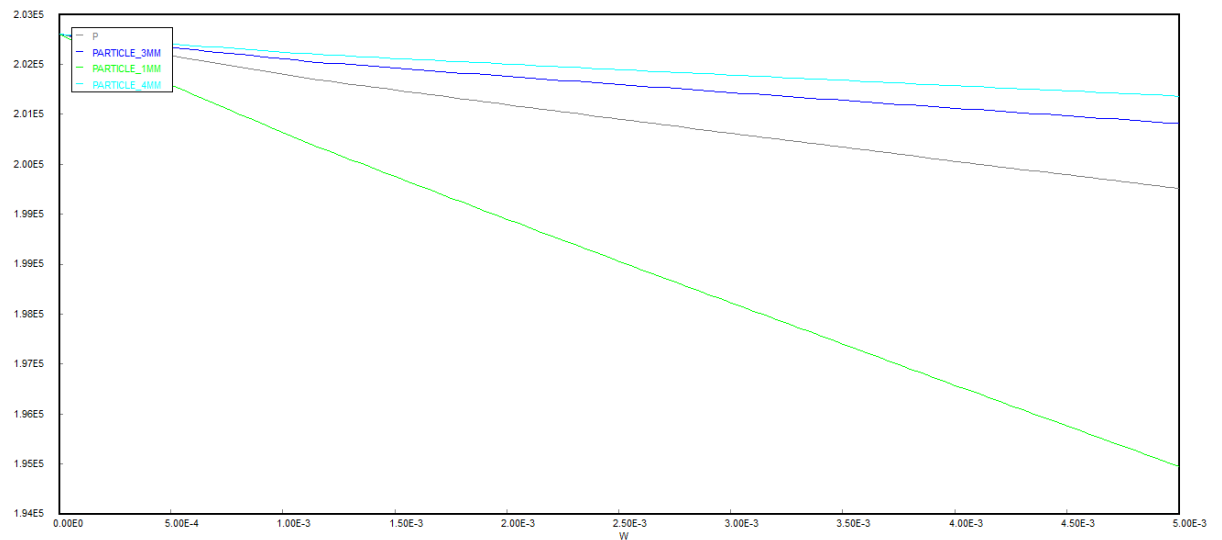
As co-current data is seen based on flow rates, pressure and reactor & salt temperatures what is important to note is that this reaction is exothermic. From analysis conversion may then be concluded to be related by each parameter co-dependently. A co-current conversion of ~59% is substantial but to an unkeen eye it is not to be defined as the best flow type. As seen in the counter current data temperature behaves in the opposite manner, mainly for the inlet salt temperature "Ta" compared to reactor temperature "T". Interestingly, the flow rates and pressure drops roughly remain constant through both flow types. To best determine the type of flow one must determine what will be used...a few tubes or multiple tubes for this PBR reactor. If few PBR tubes are available, then a co-current flow may result in greater conversion. Yet, if multiple tubes are available counter current flow will yield greater conversion based on the fundamental of this exothermic reaction and its relation to temperature change.

Deliverable 3)

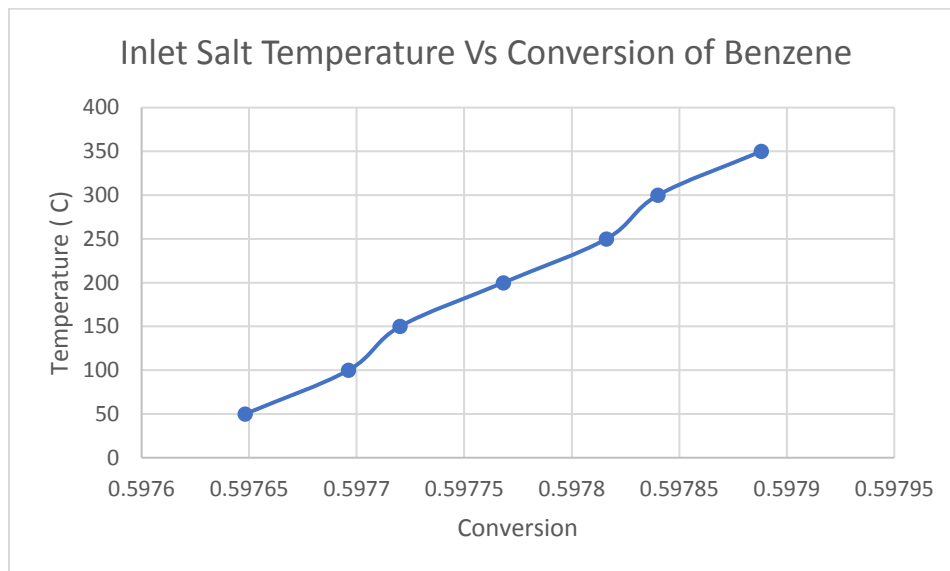
1. Plot the following

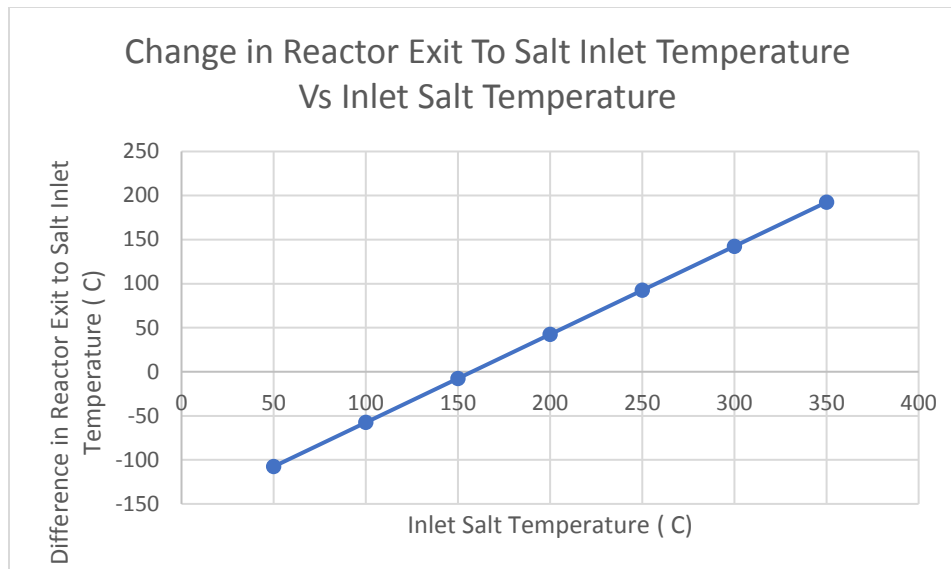
- The Pressure Profile for different particle sizes (1, 2, 3, and 4 mm)

Graph 3A: Pressure Profiles Dependent on Particle Sizes.



- Inlet salt temperature (50, 100, 150, 200, 250, 300 and 350 °C) as a function of the reactor exit temperature and final conversion (for 2 mm particle size).





Discussion Deliverable 3)

This analysis provides greater insight for the Ergun pressure drop equation. An increase in the diameter of catalyst use will cause a decrease in pressure drop. From this analysis the Ergun equation is a function of the weight of the catalyst and therefore length by the relation of dW , dV and dZ as seen beforehand. Yet, perhaps the most important factor of Deliverable 3 is the relation between inlet salt temperature and conversion. Increments of 50 C to the inlet salt temperature does increase conversion but slightly. To obtain greater conversion, as an exothermic reaction, the inlet temperature would have to increase greatly. As well as what is seen the Reactor temperature will also change in retrospect to the inlet salt. Further analysis may be seen in the relation the pressure drop differential to both reactor and salt temperatures.

Deliverable 4)

As an engineer, what changes to the design and operating parameters will you suggest increasing the conversion to 85-95%?

From the data analysis of this PBR design key parameters to change to increase a conversion to 85-95% would be parameters that would affect the differential equations. Simply looking at the Ergun pressure drop equation one may change one parameter and have an effect on the pressure drop. For it is known that change in pressure drop affects conversion. The diameter of the particle may increase in order to decrease pressure drop. In which case concentration along the PBR increases along the length of the reactor and therefore a greater conversion. To obtain the desired conversion the particle diameter size may be increased to 4+ mm with increasing the void fraction above 0.45. Also, increasing the inlet cooling temperature, the salt, will induce an increase in conversion. As seen from the data a salt inlet temperature would have to go above 400 C; and not past prime conversion temperature. As a good engineer knows, a steady and slow hand at varying parameters will help determine the perfect values for parameters to obtain desired results.

Deliverable 5)

What safety concerns are there for this reaction? How could you design your reactor to account for these concerns? For example, you might consider the effects of varying the salt inlet temperature on the resulting temperature profiles

As mentioned, a good engineer changes parameters with ease, caution, and with a slow and steady hand. Gradual changes are approved to help minimize any sudden increases in pressure, temperature and even flow rates. For example, the variation of inlet salt temperatures are with in increments of 50. If increments were to be increased to 100-200 a great change in pressure would occur. As this design is based on an aromatic such as Benzene, gases do expand quickly and rapidly. If the gas were to expand in the nearby presence of a heat source, such as the increased temperature of the salt, an explosion may occur. An explosion may not be directly with the salt, but rather with contaminants that are roaming as aerosols. To help counter act the increases in temperature the void fraction may be decreased as well to help combat these negative effects.

Closing comments)

This design is based on the parameters given and may be changed to obtain desired conversion of Benzene to Maleic Acid. Although the maximum length of the PBR was kept at 12 feet one may desire to vary the length and obtain different conversion among the length. What would be interesting to design next is varying inlet flow rates. Since this design was based on constants one may now have a greater idea on how the process may proceed. This model has unique for it does emphasize basic understanding of a chemical packed bed reactor it allows room for modifications. If cost is of interest comparisons between a PBR with fewer longer tubes to a PBR with more tubes at shorter lengths may be done so and understood. Overall, the fundamentals of chemical engineering is for greater conversion at less cost and “jumps” in parameter variations. After this design report it may be possible to do.

Appendix i.

Conversions:

$$k_1 = \frac{1E - 6 \text{ kmol}}{\text{kg.s.atm}^{0.35}} * \frac{1 \text{ atm}^{0.54}}{101325^{0.35} \left(\frac{N}{m^2}\right)^{0.35}} * \frac{1000 \text{ mol}}{1 \text{ kmol}}$$

$$k_1 = 1.77E - 5 \frac{\text{mol}}{\text{kg.s}} * \left(\frac{m^2}{N}\right)^{0.35}$$

For k₂:

$$k_2 = \frac{1.5E - 7 \text{ kmol}}{\text{kg.s.atm}} * \frac{1 \text{ atm}}{101325 \text{ N/m}^2} * \frac{1000 \text{ mol}}{1 \text{ kmol}}$$

$$k_2 = 2.66E - 6 \frac{\text{mol}}{\text{kg.s}} * \left(\frac{m^2}{N}\right)^{0.35}$$

For k₃:

$$k_3 = \frac{2.9E - 6 \text{ kmol}}{\text{kg.s.atm}^{0.35}} * \frac{1 \text{ atm}^{0.35}}{101325^{0.35} \left(\frac{N}{m^2}\right)^{0.35}} * \frac{1000 \text{ mol}}{1 \text{ kmol}}$$

$$k_3 = 2.86E - 8 \frac{\text{mol}}{\text{kg.s}} * \frac{m^2}{N}$$

U outer

Since given

$$U_{outer} = 120 \frac{W}{sK} = 19200 \frac{kJ}{sK}$$

Weight “W” of catalyst in PBR:

$$\rho_b = 3.2 \text{ E3 kg/m}^3$$

$$\text{density of bed} = \frac{\text{mass of catalyst}}{\text{Volume of bed}}$$

$$\rho_b = \frac{W}{V(\text{PBR})}$$

$$\rho_b = \frac{W}{\frac{\pi}{4} * D^2(\text{inner}) * L(\text{length of PBR})}$$

$$W = \rho_b * \frac{\pi}{4} * D^2(\text{inner}) * L(\text{length of PBR})$$

Or as

$$W = A_c * Z * \rho_b$$

$$W = 3.2 \text{ E3 } \frac{\text{kg}}{\text{m}^3} * \frac{\pi}{4} * (23.65 * 10^{-3})^2 \text{ m}^2 * 10 \text{ ft} * \frac{1 \text{ m}}{3.28 \text{ ft}}$$

$$W = 4.2 \text{ kg}$$

Inner diameter of PBR:

$$D_i(\text{inner}) = D_o(\text{outer}) - 2\{\Delta r (\text{thickness})\}$$

$$D_i(\text{inner}) = 2\text{E} - 3 \text{ m} - \left(2 \left(\frac{1}{8} \text{ in} \right) * \frac{2.54 \text{ cm}}{1 \text{ in}} * \frac{1 \text{ m}}{100 \text{ cm}} \right)$$

$$D_p(\text{inner}) = 0.01865 \text{ m}$$

Superficial Mass Velocity & density of catalyst

$$\rho_c = \frac{\rho_b}{(1 - \phi)}$$

$$\rho_c = \frac{3.2E3 \frac{kg}{m^3}}{1 - 0.45}$$

$$\rho_c = 5818.1818 \frac{kg}{m^3}$$

$$u = \frac{6.5323E - 4 \frac{m^3}{s}}{\frac{\pi}{4} * D^2(inner)}$$

$$u = \frac{6.8761 * 10^{-4} \frac{m^3}{s}}{\frac{\pi}{4} * (0.01865)^2 m^2}$$

$$u = 2.5171 \frac{m}{s}$$

Initial Volumetric flowrate

$$P_o(v_o) = (F_{To})(R)(T_o)$$

$$v_o = \frac{F_{To}(R)T_o}{P_o}$$

$$v_o = 2.77E - 05 \frac{mol}{s} * 8.314 \frac{J}{mol \cdot K} * 573.15 K * \frac{1}{2 atm} * \frac{1 atm}{101325 \frac{N}{m^2}} * \frac{N \cdot m}{1 J}$$

$$v_o = 6.5E - 7 \frac{m^3}{s}$$

Calculation for Density of Air at Inlet:

Using Ideal Gas Law

$$PV = nRT$$

$$PV = \frac{m(\text{air})}{MW(\text{air})}RT$$

$$P * MW(\text{air}) = \frac{m(\text{air})}{V}RT$$

$$P * MW(\text{air}) = \rho(\text{air})RT$$

$$\rho_o(\text{air}) = P_o * \frac{MW(\text{air})}{R * T_o}$$

$$\rho_o(\text{air}) = 1.9 \text{ atm} * \frac{29 \frac{\text{g}}{\text{mol}}}{0.0821 \text{ atm} \frac{\text{L}}{\text{mol K}}} * \frac{1}{573.15 \text{ K}} * \frac{1000 \text{ L}}{1 \text{ m}^3} * \frac{1 \text{ kg}}{1000 \text{ g}}$$

$$\rho_o(\text{air}) = 1.17096 \frac{\text{kg}}{\text{m}^3}$$

Where outer = 4/D (outer diameter)

Appendix ii:

Report for co-current flow:

POLYMATH Report

Ordinary Differential Equations

02-May-2020

Calculated values of DEQ variables

	Variable	Initial value	Minimal value	Maximal value	Final value
1	B0	2.226E+05	4.905E+04	2.226E+05	4.905E+04
2	BIGK	2.57E+33	5.263E-108	2.57E+33	5.263E-108
3	C1	1.653E+04	6143.223	1.653E+04	6143.223
4	C2	2.244887	2.244887	3.815759	3.815759
5	cp_acidc	123.9255	123.9255	178.503	178.503
6	cp_benzea	152.1785	152.1785	232.4678	232.4678
7	cp_co2e	46.2284	46.2284	58.27445	58.27445
8	cp_cof	30.23817	30.23817	32.9479	32.94739
9	cp_nitrog	29.95785	29.95785	32.84849	32.84812
10	cp_oxygenb	31.79986	31.79986	34.58704	33.54736
11	cp_salt	50.72168	44.48089	50.72168	44.48089
12	cp_waterd	35.50431	32.74251	37.39102	32.74251
13	deltacp	-6.794086	-36.20258	-6.794086	-36.19878
14	density	4.2095	1.564152	4.2095	1.564152
15	density_air_initial	1.17	1.17	1.17	1.17
16	Facidc	0	0	7.9E-06	3.466E-06
17	Fbenzea	0.004167	0.0016754	0.004167	0.0016754
18	Fco2e	0	0	0.0106106	0.0106106
19	Fcof	0	-4.299E-49	0.0043252	0.0043252
20	Fnitrog	0.01181	0.01181	0.01181	0.01181
21	Foxygenb	0.01181	-0.0047041	0.01181	-0.0047041
22	Ftotal	0.027787	0.0277839	0.031192	0.031192
23	Fwaterd	0	0	0.0074714	0.0074714
24	G	6.409712	2.381699	6.409712	2.381699
25	h1rxn	-1.236E+06	-1.236E+06	-1.236E+06	-1.236E+06
26	h2rxn	-2.656E+06	-2.656E+06	-2.656E+06	-2.656E+06
27	h3rxn	-2.656E+06	-2.656E+06	-2.656E+06	-2.656E+06
28	int_acidc	3.731E+04	3.731E+04	1.629E+05	1.629E+05
29	int_co2e	2.027E+04	2.027E+04	6.242E+04	6.242E+04
30	int_cof	1.697E+04	1.697E+04	4.261E+04	4.261E+04
31	int_delcp1	-671.1177	-1.363E+04	-671.1177	-1.363E+04

32	int_delcp2	1.191E+04	-9884.202	1.191E+04	-9884.202
33	int_delcp3	2.285E+04	7821.714	2.286E+04	7821.714
34	int_oxygen	1.687E+04	1.687E+04	4.37E+04	4.37E+04
35	int_oxygenenz	4.144E+04	4.144E+04	2.008E+05	2.008E+05
36	int_waterd	1.942E+04	1.942E+04	4.804E+04	4.804E+04
37	K1	1.331E-06	1.331E-06	0.0323428	0.0323428
38	K2	2.001E-07	2.001E-07	0.0048605	0.0048605
39	K3	2.293E-10	2.293E-10	0.0347021	0.0347021
40	P	2.027E+05	2.008E+05	2.027E+05	2.008E+05
41	P_ACID	3.04E+04	1.211E+04	3.04E+04	1.211E+04
42	P_BENZ	0	0	57.40989	25.05409
43	r1	4.935E-05	1.993E-22	1.191907	0.8687127
44	r2	7.417E-06	7.417E-06	0.1791227	0.1305523
45	r3	0	0	1.19245	0.8694304
46	sum_Fbenzeandcp	1.363486	1.363486	1.750504	1.625682
47	T	573.15	573.15	1361.536	1361.536
48	Ta	298.15	298.15	298.1507	298.1507
49	U	1.522678	1.522678	1.522678	1.522678
50	viscos	5.442E-05	5.442E-05	9.25E-05	9.25E-05
51	W	0	0	0.0032	0.0032
52	z	1.936E+05	-8020.259	1.202E+07	-6172.663

Differential equations

- 1 $d(\text{Fbenzea})/d(W) = (-r_1 - r_2)$
- 2 $d(\text{Foxygenb})/d(W) = ((-4.5 * r_1) - (7.5 * r_2) - (2 * r_3))$
- 3 $d(\text{Facidc})/d(W) = (r_1 - r_3)$
- 4 $d(\text{Fwaterd})/d(W) = ((2 * r_1) + (3 * r_2) + r_3)$
- 5 $d(\text{Fnitrog})/d(W) = 0$
- 6 $d(\text{Fcof})/d(W) = 2 * r_3$
- 7 $d(\text{Fco2e})/d(W) = ((2 * r_1) + (6 * r_2) + (2 * r_3))$
- 8 $d(P)/d(W) = (-B_0 / ((3.2 * 10^3) * (1 - 0.45) * (1 - 0.45) * (2.73 * (10^{-4})))) * (202650/P) * (T/573.15) * (F_{\text{total}}/0.02778))$
- 9 $d(T_a)/d(W) = (6 * (T - T_a) / ((c_{p_salt}) * 555.56))$
- 10 $d(T)/d(W) = ((-6 * (T_a - T) + z) / \text{sum_Fbenzeandcp})$

Explicit equations

- 1 $P_{\text{BENZ}} = (\text{Facidc}/0.02778) * P$
- 2 $P_{\text{ACID}} = (\text{Fbenzea}/0.02778) * P$
- 3 $c_{p_co2e} = 4.184 * (4.7 + ((17.4 * 10^{-3}) * T) + ((-13.38 * 10^{-6}) * (T^2)) + ((4.097 * 10^{-9}) * (T^3)))$
- 4 $K_1 = (1.77 * 10^{-5}) * (\exp((83100/8.314) * ((1/673) - (1/T))))$


```

5 K2 = (2.66*10^-6)*(exp((83100/8.314)*((1/673)-(1/T))))
6 K3 = (2.86*10^-8)*(exp((155000/8.314)*((1/673)-(1/T))))
7 cp_oxygenb = 4.184*(6.71+((-0.0009*10^-3)*T)+((4.17*10^-6)*(T^2))+((-2.544*10^-9)*(T^3)))
8 r2 = K2*(P_ACID^0.35)
9 cp_waterd = 4.184*(8.05+((-1.67*10^-3)*T)+((6.519*10^-6)*(T^2))+((-3.976*10^-9)*(T^3)))
10 h1rxn = -1236*(10^3)
11 h2rxn = -2656*(10^3)
12 h3rxn = -2656*(10^3)
13 cp_benzea = 4.184*(-8.1010+((113.3000*10^-3)*T)+((-72.0600*10^-6)*(T^2))+((17.0300*10^-9)*(T^3)))
14 deltacp = (6*cp_co2e)+(3*cp_waterd)-cp_benzea-(7.5*cp_oxygenb)
15 cp_acidc = 4.184*(-3.123+((83.23*10^-3)*T)+((-52.17*10^-6)*(T^2))+((11.56*10^-9)*(T^3)))
16 BIGK = 0.0031*exp((( -2656*10^3)-(573.15*deltacp))/(8.314)*((1/673)-(1/T))+((deltacp/8.314)*ln(T/673)))
17 r1 = K1*(P_ACID^0.35)/(1+BIGK*(P_BENZ^2))
18 cp_cof = 4.184*(7.37+((-3.07*10^-3)*T)+((6.662*10^-6)*(T^2))+((-3.037*10^-9)*(T^3)))
19 cp_nitrog = 4.184*(7.44+((-3.24*10^-3)*T)+((6.4*10^-6)*(T^2))+((-2.79*10^-9)*(T^3)))
20 cp_salt = 50+((2*10^-3)*T)+((1*10^-6)*(T^2))+((-4*10^-9)*(T^3))
21 r3 = K3*P_BENZ/((1+(BIGK*P_BENZ))^2)
22 int_oxygenenz = 4.184*((-8.1*(T-0))+(((113.3*10^-3)/2)*((T^2)-(0^2)))+((( -72.06*10^-6)/3)*((T^3)-(0^3)))+(((17.03*10^-9)/4)*((T^4)-(0^4))))
23 int_oxygen = 4.184*((6.7*(T-0))+((( -0.0009*10^-3)/2)*((T^2)-(0^2)))+(((4.17*10^-6)/3)*((T^3)-(0^3)))+((( -2.544*10^-9)/4)*((T^4)-(0^4))))
24 int_acidc = 4.184*((-3.123*(T-0))+(((83.23*10^-3)/2)*((T^2)-(0^2)))+((( -52.17*10^-6)/3)*((T^3)-(0^3)))+(((11.56*10^-9)/4)*((T^4)-(0^4))))
25 int_waterd = 4.184*((8.05*(T-0))+((-1.67*10^-3)/2)*((T^2)-(0^2)))+(((6.519*10^-6)/3)*((T^3)-(0^3)))+((-3.976*10^-9)/4)*((T^4)-(0^4)))
26 int_co2e = 4.184*((4.7*(T-0))+(((17.54*10^-3)/2)*((T^2)-(0^2)))+((-13.38*10^-6)/3)*((T^3)-(0^3)))+(((4.097*10^-9)/4)*((T^4)-(0^4)))
27 int_cof = 4.184*((7.37*(T-0))+((-3.07*10^-3)/2)*((T^2)-(0^2)))+(((6.662*10^-6)/3)*((T^3)-(0^3)))+((-3.037*10^-9)/4)*((T^4)-(0^4)))
28 int_delp1 = (2*int_waterd)+(int_acidc)+(2*int_co2e)-(int_oxygenenz)-(4.5*int_oxygen)

```

```

2
9 int_delcp2 = (3*int_waterd)+(6*int_co2e)-(int_oxygenenz)-(7.5*int_oxygen)

3
0 int_delcp3 = (int_waterd)+(2*int_co2e)+(2*int_cof)-(int_acidc)-(2*int_oxygen)

3
1 z = -r1*((h1rxn)*(int_delcp1))-r2*((h2rxn)*(int_delcp2))-r3*((h3rxn)*(int_delcp3))

3
2 sum_Fbenzeandcp =
(Fbenzea*cp_benzea)+(Foxygenb*cp_oxygenb)+(Facidc*cp_acidc)+(Fwaterd*cp_waterd)+(Fco2e*cp_co2e)+(F
cof*cp_cof)+(Fnitrog*cp_nitrog)

3
3 Ftotal = Fbenzea+Foxygenb+Facidc+Fwaterd+Fco2e+Fcof+Fnitrog

3
4 density_air_initial = 1.17

3
5 viscos = 1.7894*(10^-5)*(((T/273))^(1.5))*((573+110)/(T+110))

3
6 C2 = ((150*(1-0.45)*viscos)/(2*10^-3))

3
7 density = density_air_initial*(P/202650)*(573/T)*(0.1/Ftotal)

3
8 U = ((6.6876*10^-4)/(4.392*10^-4))

3
9 G = density*U

4
0 C1 = (G*(1-0.45))/((density_air_initial*1*(2*10^-3)*(0.45^3))

4
1 B0 = C1*(C2+(1.75*G))

```

General

Total number of equations	51
Number of differential equations	10
Number of explicit equations	41
Elapsed time	21.991 sec
Solution method	RKF_45
Step size guess. h	0.000001
Truncation error tolerance. eps	0.000001

Appendix iii:

Poymath Code:

#Definitive values

$$P_BENZ = (Facidc / 0.02778) * P$$

$$P_ACID = (Fbenzea / 0.02778) * P$$

$$BIGK = 0.0031 * \exp(((-2656 * 10^3) - (573.15 * \text{deltacp})) / (8.314) * ((1/673) - (1/T)) + ((\text{deltacp} / 8.314) * \ln(T/673)))$$

$$K1 = (1.77 * 10^{-5}) * (\exp((83100 / 8.314) * ((1/673) - (1/T))))$$

$$K2 = (2.66 * 10^{-6}) * (\exp((83100 / 8.314) * ((1/673) - (1/T))))$$

$$K3 = (2.86 * 10^{-8}) * (\exp((155000 / 8.314) * ((1/673) - (1/T))))$$

$$r1 = K1 * (P_ACID^{0.35}) / (1 + BIGK * (P_BENZ^2))$$

$$r2 = K2 * (P_ACID^{0.35})$$

$$r3 = K3 * P_BENZ / ((1 + (BIGK * P_BENZ))^2)$$

$$h1rxn = -1236 * (10^3)$$

$$h2rxn = -2656 * (10^3)$$

$$h3rxn = -2656 * (10^3)$$

#Differential Flow rates

$$W(0) = 0$$

$$W(f) = 0.0032$$

$$d(Fbenzea) / d(W) = (-r1 - r2)$$

$$Fbenzea(0) = 0.004167$$

$$d(Foxygenb) / d(W) = ((-4.5 * r1) - (7.5 * r2) - (2 * r3))$$

$$Foxygenb(0) = 0.01181$$

$$d(Facidc) / d(W) = (r1 - r3)$$

$$Facidc(0) = 0$$

$$d(Fwaterd) / d(W) = ((2 * r1) + (3 * r2) + r3)$$

$$Fwaterd(0) = 0$$

$$d(Fnitrog) / d(W) = 0$$

$$Fnitrog(0) = 0.01181$$

$$d(F_{cof}) / d(W) = 2*r3$$

$$F_{cof}(0) = 0$$

$$d(F_{co2e}) / d(W) = (2*r1) + (6*r2) + (2*r3)$$

$$F_{co2e}(0) = 0$$

#differential Pressure Drop

$$d(P) / d(W) = (-B0 / ((3.2*10^{-3}) * (1-0.45) * (1-0.45) * (2.73*(10^{-4}))) * (202650/P) * (T/573.15) * (F_{total}/0.02778))$$

$$P(0) = 202650$$

#Differential Temperatures

$$d(T_a) / d(W) = (6*(T-T_a) / ((c_{p_salt}) * 555.56))$$

$$T_a(0) = 298.15$$

$$d(T) / d(W) = ((-6*(T_a-T) + z) / \text{sum_Fbenzeandcp})$$

$$T(0) = 573.15$$

***for counter current "Ta-T" becomes "T-Ta"

#Cp

$$c_{p_benzea} = 4.184 * (-8.1010 + ((113.3000*10^{-3})*T) + ((-72.0600*10^{-6})*(T^2)) + ((17.0300*10^{-9})*(T^3)))$$

$$c_{p_oxygenb} = 4.184 * (6.71 + ((-0.0009*10^{-3})*T) + ((4.17*10^{-6})*(T^2)) + ((-2.544*10^{-9})*(T^3)))$$

$$c_{p_acidc} = 4.184 * (-3.123 + ((83.23*10^{-3})*T) + ((-52.17*10^{-6})*(T^2)) + ((11.56*10^{-9})*(T^3)))$$

$$c_{p_waterd} = 4.184 * (8.05 + ((-1.67*10^{-3})*T) + ((6.519*10^{-6})*(T^2)) + ((-3.976*10^{-9})*(T^3)))$$

$$c_{p_co2e} = 4.184 * (4.7 + ((17.4*10^{-3})*T) + ((-13.38*10^{-6})*(T^2)) + ((4.097*10^{-9})*(T^3)))$$

$$c_{p_cof} = 4.184 * (7.37 + ((-3.07*10^{-3})*T) + ((6.662*10^{-6})*(T^2)) + ((-3.037*10^{-9})*(T^3)))$$

$$c_{p_nitrog} = 4.184 * (7.44 + ((-3.24*10^{-3})*T) + ((6.4*10^{-6})*(T^2)) + ((-2.79*10^{-9})*(T^3)))$$

$$c_{p_salt} = 50 + ((2*10^{-3})*T) + ((1*10^{-6})*(T^2)) + ((-4*10^{-9})*(T^3))$$

$$\text{deltacp} = (6*c_{p_co2e}) + (3*c_{p_waterd}) - c_{p_benzea} - (7.5*c_{p_oxygenb})$$

#Cp Integrals

$$\text{int_oxygenenz} = 4.184 * ((-8.1*(T-0)) + (((113.3*10^{-3})/2)*((T^2)-(0^2))) + (((-72.06*10^{-6})/3)*((T^3)-(0^3))) + (((17.03*10^{-9})/4)*((T^4)-(0^4))))$$

```

int_oxygen=4.184*((6.7*(T-0))+((-0.0009*10^-3)/2)*((T^2)-(0^2))+((4.17*10^-6)/3)*((T^3)-(0^3))+((-
2.544*10^-9)/4)*((T^4)-(0^4)))

int_acidc=4.184*((-3.123*(T-0))+((83.23*10^-3)/2)*((T^2)-(0^2))+((-52.17*10^-6)/3)*((T^3)-
(0^3))+((11.56*10^-9)/4)*((T^4)-(0^4)))

int_waterd=4.184*((8.05*(T-0))+((-1.67*10^-3)/2)*((T^2)-(0^2))+((6.519*10^-6)/3)*((T^3)-(0^3))+((-
3.976*10^-9)/4)*((T^4)-(0^4)))

int_co2e=4.184*((4.7*(T-0))+((17.54*10^-3)/2)*((T^2)-(0^2))+((-13.38*10^-6)/3)*((T^3)-(0^3))+((4.097*10^-
9)/4)*((T^4)-(0^4)))

int_cof=4.184*((7.37*(T-0))+((-3.07*10^-3)/2)*((T^2)-(0^2))+((6.662*10^-6)/3)*((T^3)-(0^3))+((-3.037*10^-
9)/4)*((T^4)-(0^4)))

int_delcp1=(2*int_waterd)+(int_acidc)+(2*int_co2e)-(int_oxygenenz)-(4.5*int_oxygen)

int_delcp2=(3*int_waterd)+(6*int_co2e)-(int_oxygenenz)-(7.5*int_oxygen)

int_delcp3=(int_waterd)+(2*int_co2e)+(2*int_cof)-(int_acidc)-(2*int_oxygen)

#length param

z=-r1*((h1rxn)*(int_delcp1))-r2*((h2rxn)*(int_delcp2))-r3*((h3rxn)*(int_delcp3))

#basics

sum_Fbenzeandcp=(Fbenzea*cp_benzea)+(Foxygenb*cp_oxygenb)+(Facidc*cp_acidc)+(Fwaterd*cp_waterd)+(Fco
2e*cp_co2e)+(Fcof*cp_cof)+(Fnitrog*cp_nitrog)

Ftotal=Fbenzea+Foxygenb+Facidc+Fwaterd+Fco2e+Fcof+Fnitrog

#for ergun pressure drop

density_air_initial=1.17

viscos=1.7894*(10^-5)*(((T/273))^(1.5))*((573+110)/(T+110))

C2=((150*(1-0.45)*viscos)/(2*10^-3))

density=density_air_initial*(P/202650)*(573/T)*(0.1/Ftotal)

U=((6.6876*10^-4)/(4.392*10^-4))

G=density*U

C1=(G*(1-0.45))/(density_air_initial*1*(2*10^-3)*(0.45^3))

B0=C1*(C2+(1.75*G))

```

